

Digital Transmission

Physical

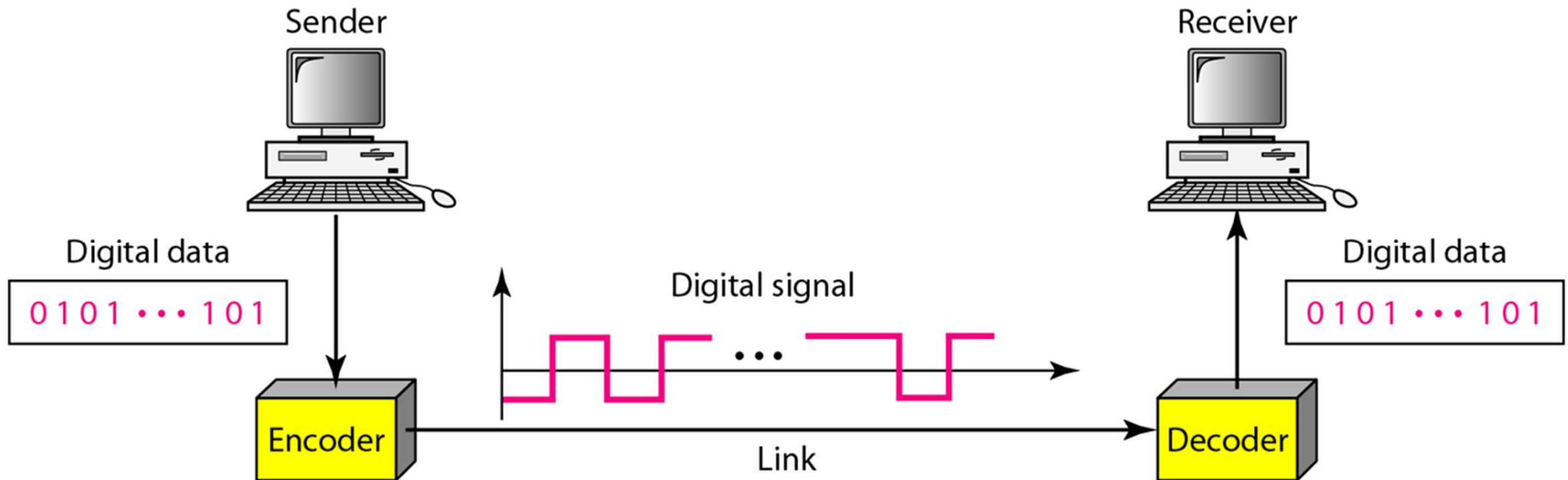
Outline

- Digital-to-Digital Conversion
- Analog-to-Digital Conversion

Digital-to-Digital Conversion

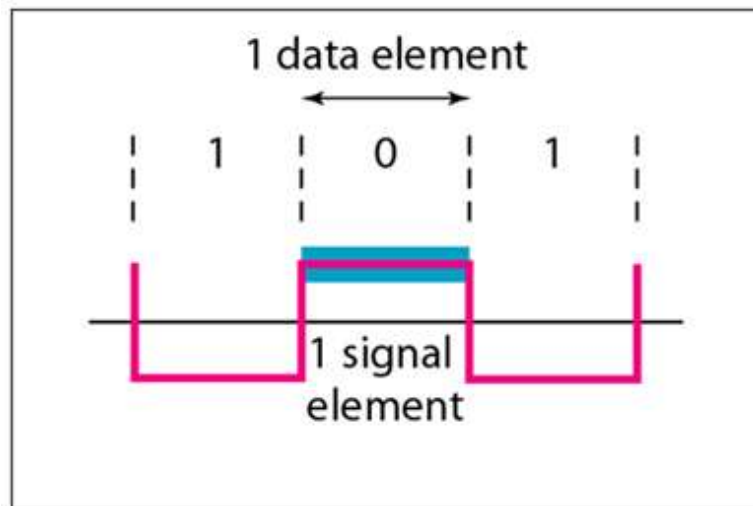
Line Coding

- Convert binary data into digital signals



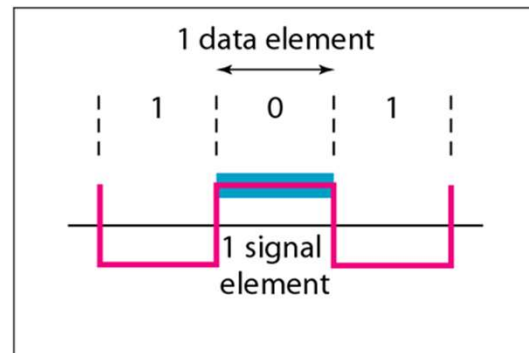
Data Element and Signal Element

- Data element
 - unit of digital data = bit
- Signal element
 - unit of digital signal

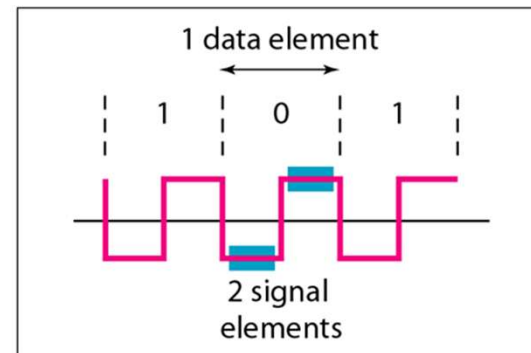


Data Element and Signal Element

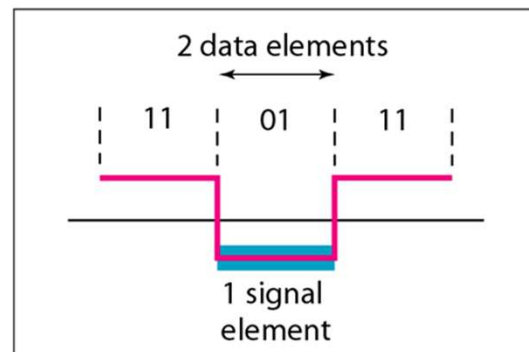
- r = number of data elements per signal element
 - bits per symbol



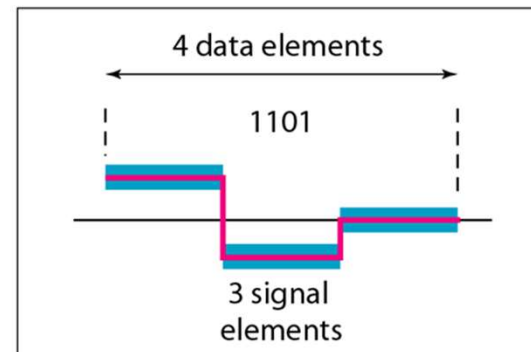
a. One data element per one signal element ($r = 1$)



b. One data element per two signal elements ($r = \frac{1}{2}$)



c. Two data elements per one signal element ($r = 2$)



d. Four data elements per three signal elements ($r = \frac{4}{3}$)

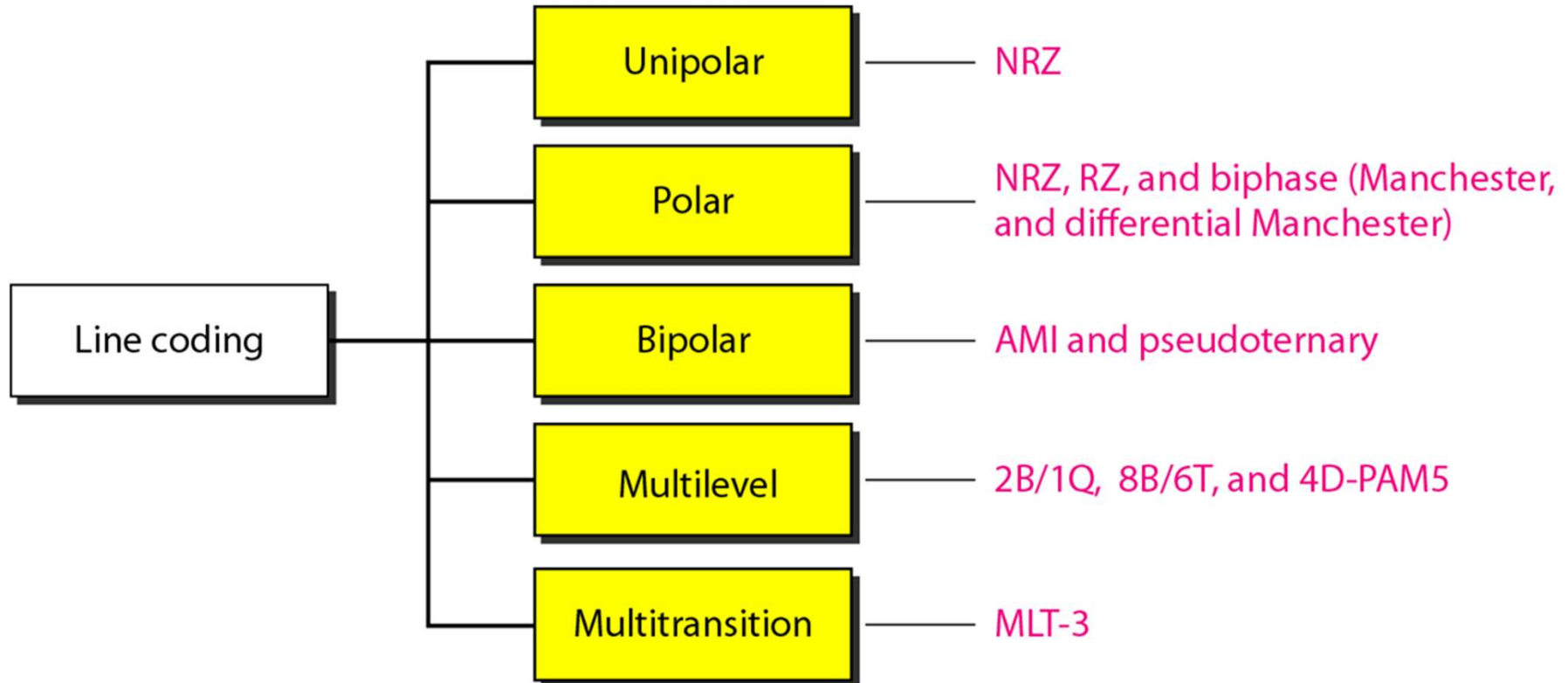
Units of Data and Signal Transmission

- Data transmission rate
 - number of data elements sent in a second
 - unit: bps
- Signal transmission rate
 - number of signal elements sent in a second
 - unit: baud

Exercise

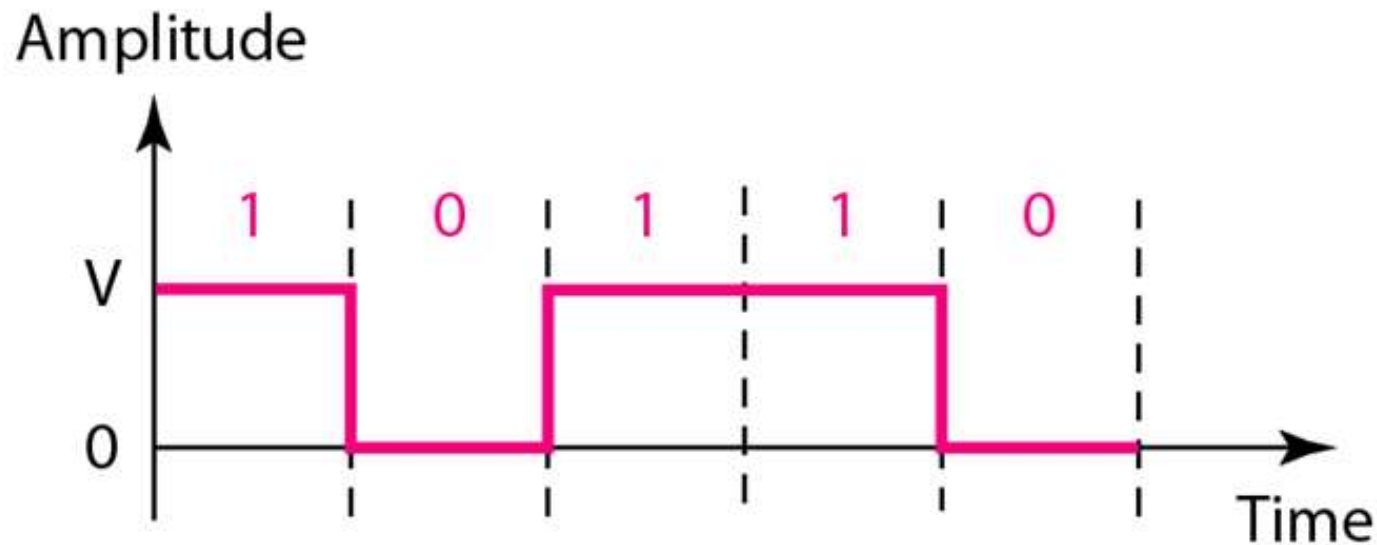
- A signal is carrying data in which one data element is encoded as two signal element. If the bit rate is 100kbps, what is the baud rate?

Line Coding Schemes



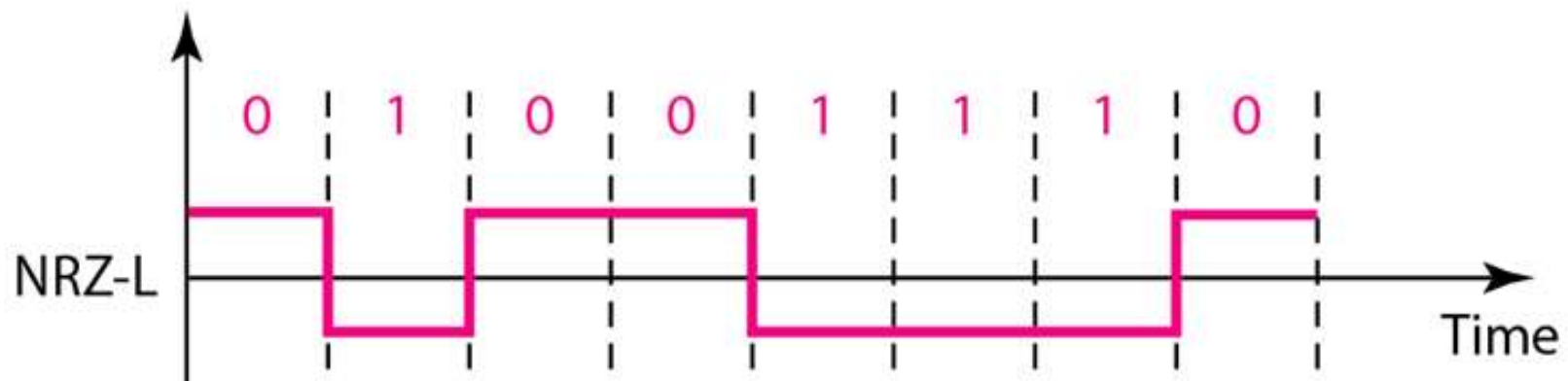
Unipolar Encoding

- Uses only one polarity: signal levels are 0V and +V (no negative voltage)
- Unipolar NRZ (non-return-to-zero)
 - Bit 1 → high volt
 - Bit 0 → zero volt



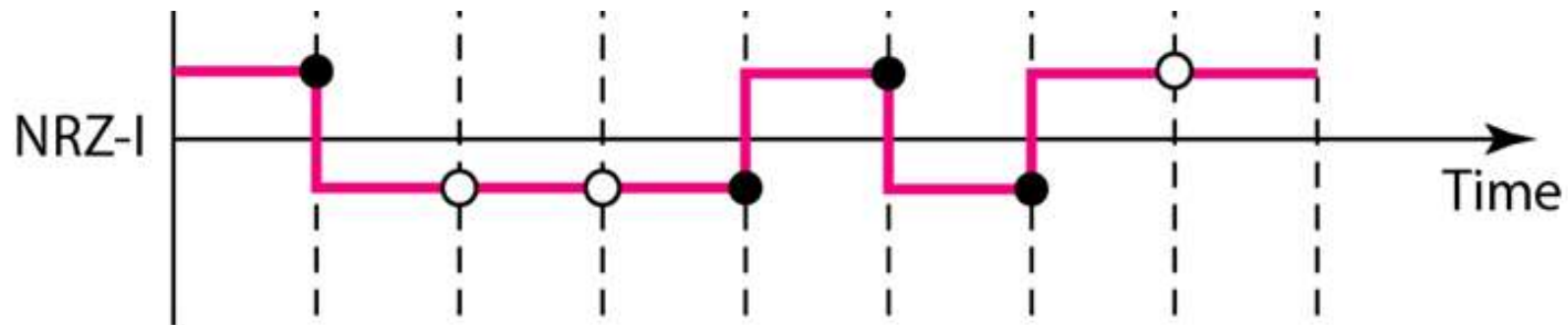
Polar Encoding

- Uses both +V and -V (two polarities)
- NRZ-L (Level): voltage level determines bit value
 - Bit 0 → +V, Bit 1 → -V



Polar Encoding

- NRZ-I (Inverted): transition at the beginning determines bit value
 - Bit 1 → transition
 - Bit 0 → no transition



DC Component

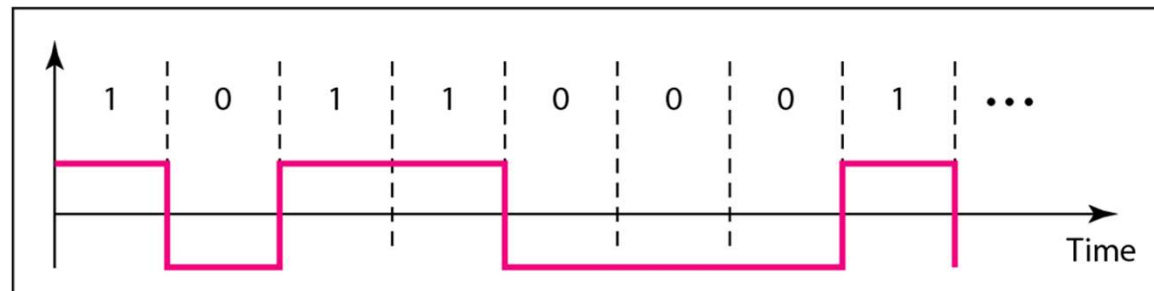
- DC component: zero-frequency component in the frequency domain
 - The average value of the signal is nonzero
- NRZ schemes can produce DC component, especially when the same bit value continues for a long time
- Problem: DC component cannot pass through AC-coupled links (e.g. transformers, capacitors)
- Two consequences
 - Baseline wandering
 - Clock drift

Baseline Wandering

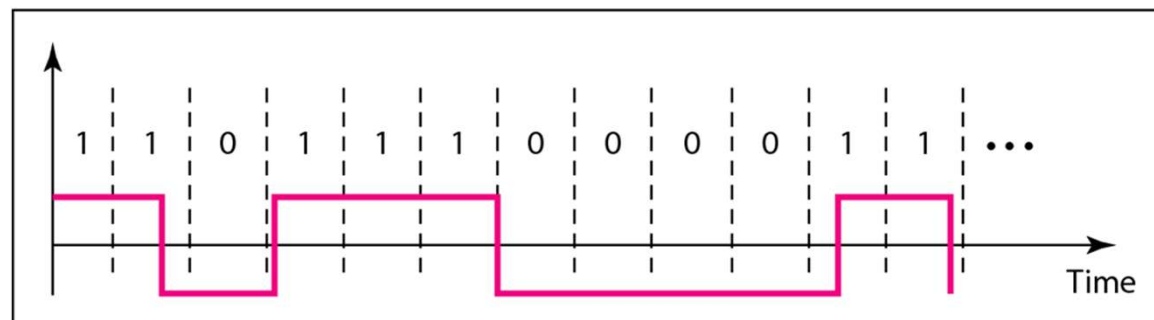
- AC-coupled links (transformers, capacitors) can only pass changing signals - not constant levels
- When the same bit continues for a long time, the signal droops toward 0V at the receiver
 - The DC level of the received signal shifts over time - baseline wandering
- As the signal level approaches 0V, reducing the noise margin
 - Small noise can cause the receiver to misinterpret the bit value

Clock Drift

- The speed of clock is slightly different for devices
- Time mismatch of sender and receiver can lead to decoding problems, since the receiver may not know when the signal element ends



a. Sent



b. Received

Self Synchronization

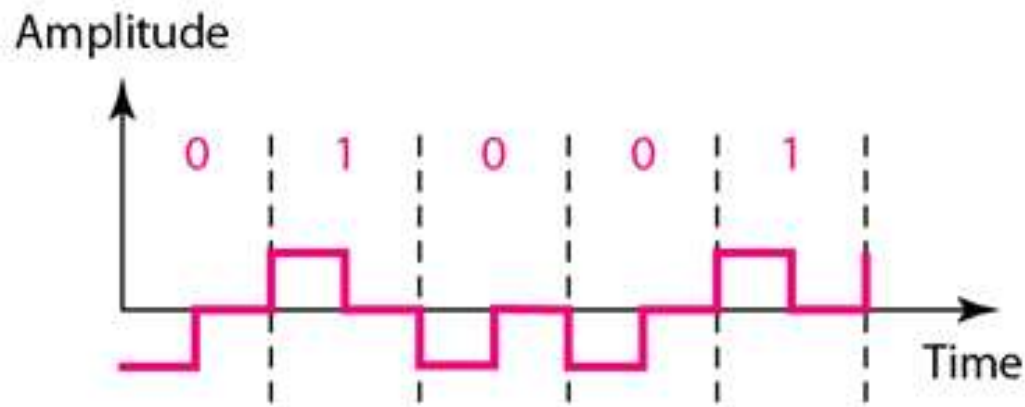
- Even if the speed of clocks is different, the receiver can synchronize clocks if signal changes frequently
- If an encoding scheme enforces signal changes, the scheme is said to have the “self synchronization” property

Self Synchronization

- NRZ-L and NRZ-I do not have self-synchronization—
Long runs of the same bit produce no transitions →
clock drift
- NRZ-I is better than NRZ-L
 - NRZ-L: long run of either 0 or 1 → no transition
 - NRZ-I: only long run of 0 causes no transition (bit 1 always produces a transition)

Polar Encoding

- RZ (Return-to-Zero)
 - Each bit uses two signal elements
 - First half: $+V$ (bit 1) or $-V$ (bit 0)
 - Second half: always returns to $0V$

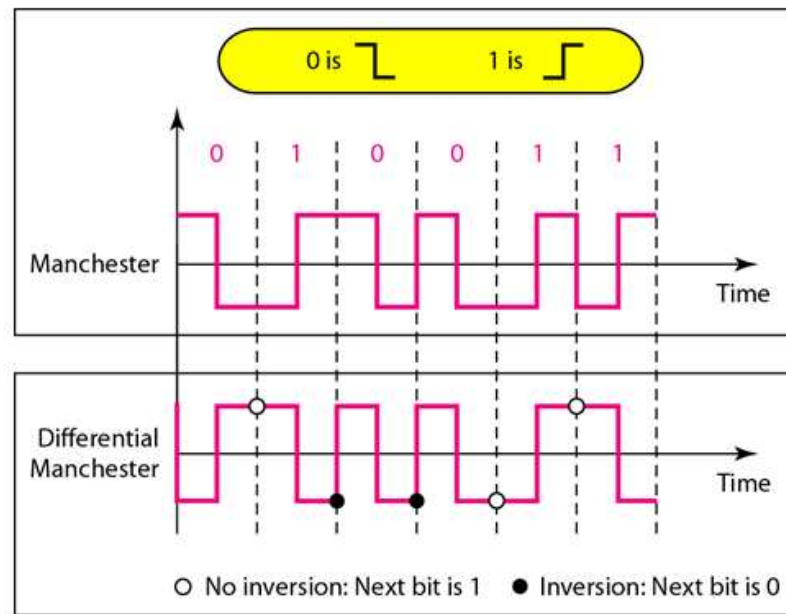


Polar Encoding

- RZ (Return-to-Zero)
- Pros: self-synchronization
- Cons
 - Two signal elements per data element
 - requires double baud rate for the same bit rate
 - Three signal levels (LOW, 0, HIGH)
 - lower noise margin
 - Baseline wandering not solved

Polar Encoding

- Manchester encoding: mid-bit transition determines bit value
 - Bit 0: $+V \rightarrow -V$ (high to low)
 - Bit 1: $-V \rightarrow +V$ (low to high)
- Differential Manchester: mid-bit transition always present; beginning bit determines value
 - Bit 0: transition at the beginning
 - Bit 1: no transition at the beginning

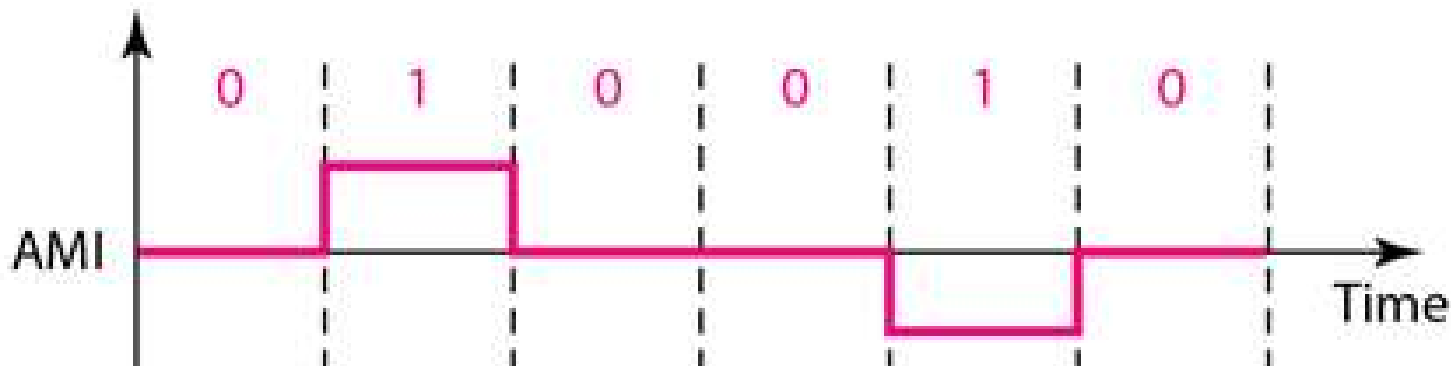


Polar Encoding

- Manchester, differential Manchester encoding
 - Compared to RZ, fewer signal levels
 - Requires double baud rate (same as RZ)
 - Signal changes every bit
 - No DC component
 - No baseline wandering
 - No clock drift

Bipolar Encoding

- Uses three levels: $+V$, $0V$, $-V$
 - One bit value: $0V$
 - The other $\rightarrow$ alternates between $+V$ and $-V$
- AMI (Alternate Mark Inversion)
 - Bit 0 $\rightarrow 0V$
 - Bit 1 $\rightarrow$ Alternate $+V$ and $-V$
 - Removes DC component: $+V$ and $-V$ cancel each other out



Multilevel Encoding

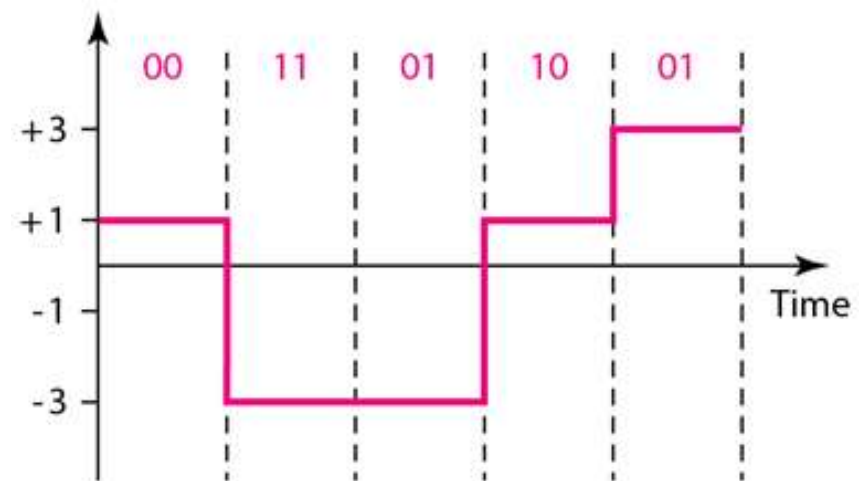
- Use multiple signal levels to encode data
- More bits per signal element → higher bit rate at the same baud rate
- mBnL: Use n signal element to encode m data elements
 - ‘L’ is number of signal levels
 - B (Binary): 2 levels
 - T (Ternary): 3 levels
 - Q (Quaternary): 4 levels
- $2^m \leq L^n$
 - All possible m-bit patterns must be representable by n signal elements

Multilevel Encoding

- 2B1Q
 - 2 bits are encoded into a single signal element
 - 4 signal levels (Q)
 - Used in DSL (Digital Subscriber Line)

Next bits	Previous level:	
	positive	negative
Next level	Next level	Next level
00	+1	-1
01	+3	-3
10	-1	+1
11	-3	+3

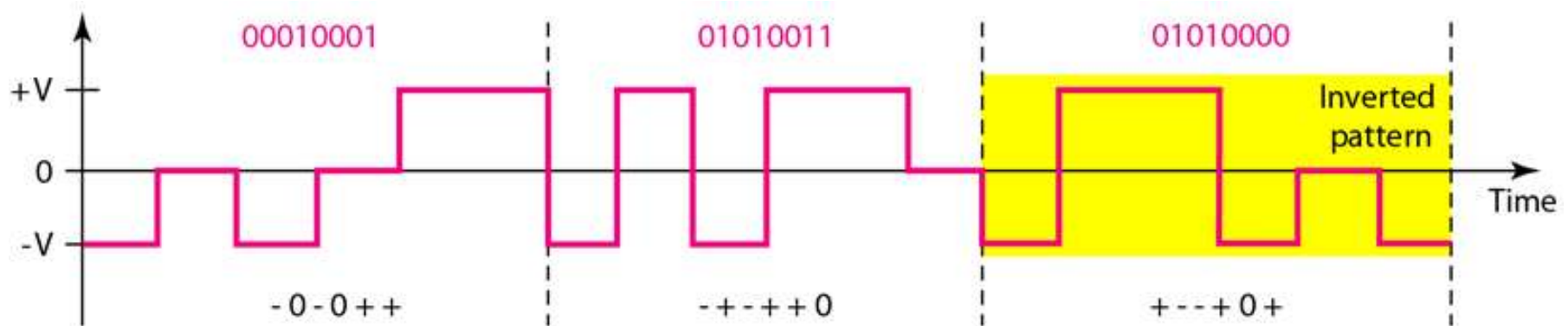
Transition table



Assuming positive original level

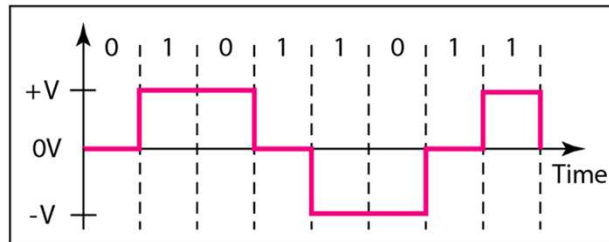
Multilevel Encoding

- 8B6T
 - Used in 100BASE-4T
 - 8 bits are encoded into 6 three-level signal element
 - 6 signal elements are designed so that
 - The element has the same number of '+'s and '-'s, or
 - The element has one more '+'s than '-'s
 - If a code has one more '+' than '-', every other occurrence of such code is inverted to maintain DC balance

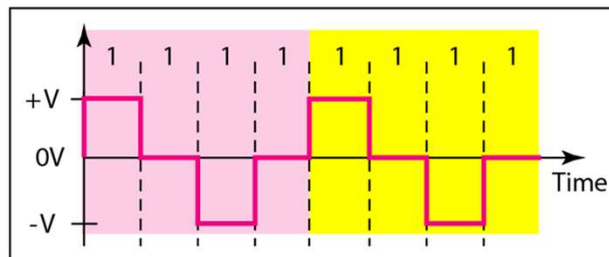


MLT-3

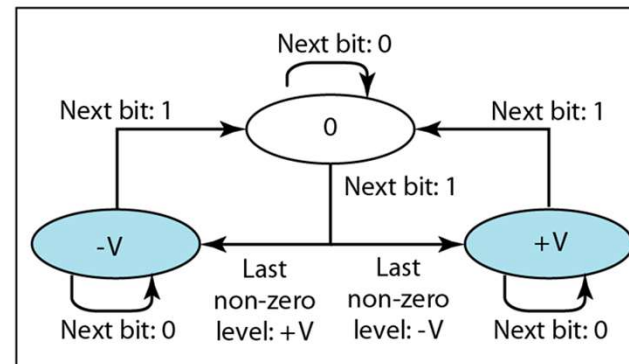
- Used in LAN
 - If the next bit is 0, no signal level change
 - If the next bit is 1 and the current signal level is nonzero, then signal level is 0
 - If the next bit is 1 and the current signal level is 0, then signal level is the inverse of most recent nonzero level



a. Typical case



b. Worse case



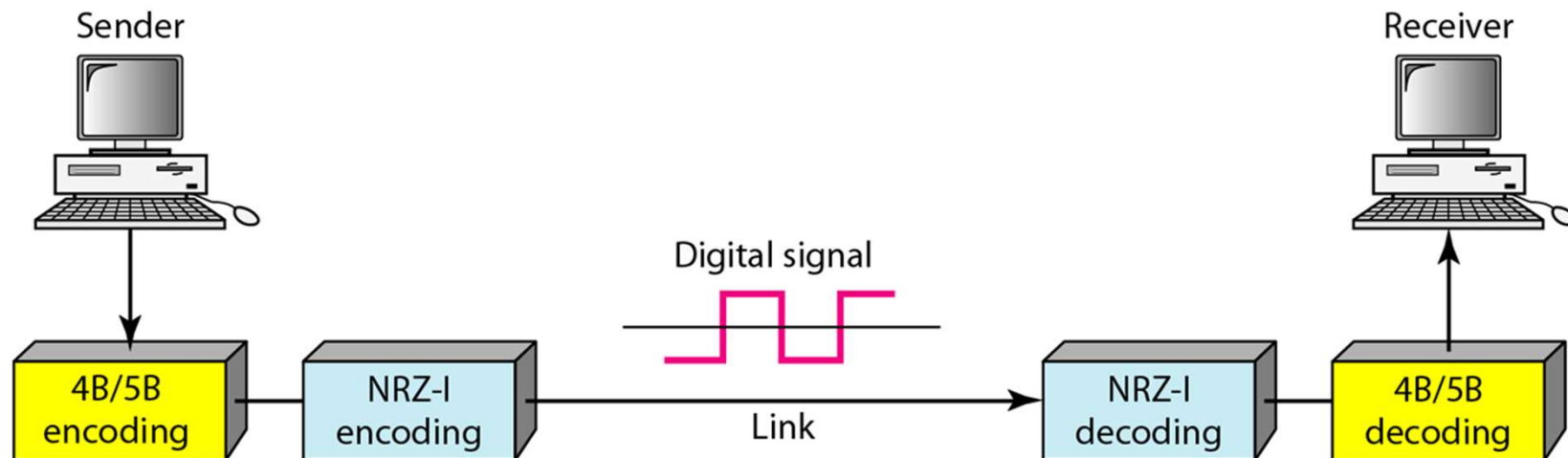
c. Transition states

Block Coding

- Convert m bits of data into n bits of data (block code)
- called mB/nB coding
- $m < n$
 - extra bits provide redundancy for synchronization and error detection
- Goal: self-synchronization, error detection, etc.

Block Coding

- 4B/5B
 - NRZ-I shows good transmission rate ($r = 1$), but has synchronization problem
 - If series of 0s continues, synchronization problem can occur
 - The goal of 4B/5B is to avoid continuous 0s



Block Coding

- 4B/5B
 - Convert 4-bit data into 5 bits so that
 - No more than three consecutive 0s appear in the encoded stream
 - 001001011001101000011000 (X) - contains "0000"

Block Coding

- 4B/5B
 - 4-bit data: 0000, 0001, 0010, ..., 1110, 1111
 - We need to convert 4-bit data into 5-bit block codes
 - Among 5 bits, the first 2 bits must include at least one '1', and the last 3 bits must include at least one '1'.
 - How many 5-bit codes exist, that satisfy the previous condition?
 - Is this block coding possible? How can we prove this?

Block Coding

- Example block coding

<i>Data Sequence</i>	<i>Encoded Sequence</i>	<i>Control Sequence</i>	<i>Encoded Sequence</i>
0000	11110	Q (Quiet)	00000
0001	01001	I (Idle)	11111
0010	10100	H (Halt)	00100
0011	10101	J (Start delimiter)	11000
0100	01010	K (Start delimiter)	10001
0101	01011	T (End delimiter)	01101
0110	01110	S (Set)	11001
0111	01111	R (Reset)	00111
1000	10010		
1001	10011		
1010	10110		
1011	10111		
1100	11010		
1101	11011		
1110	11100		
1111	11101		

Example

- Suppose signal is transmitted at 1M baud
- What is the data rate if NRZ-I + 4B/5B is used?
- What is the data rate if Manchester Encoding is used?

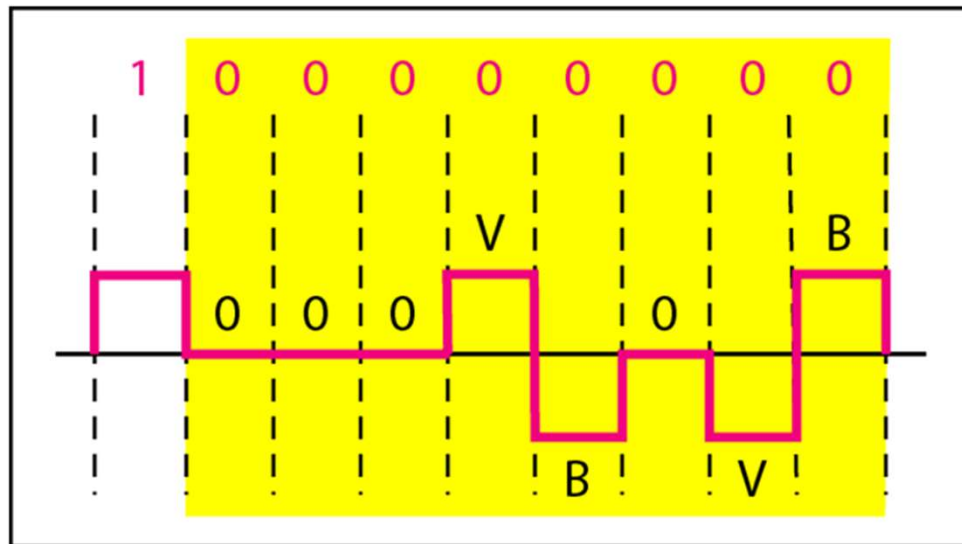
Scrambling

- AMI encoding alternates LOW and HIGH signals in order to remove DC component
- However, since bit 0 is encoded into 0V, synchronization problem can occur if a series of 0 continues
- Scrambling is a scheme that removes continuous 0 without incurring overhead
 - Unlike block coding (e.g., 4B/5B), scrambling replaces patterns in-place - no increase in baud rate

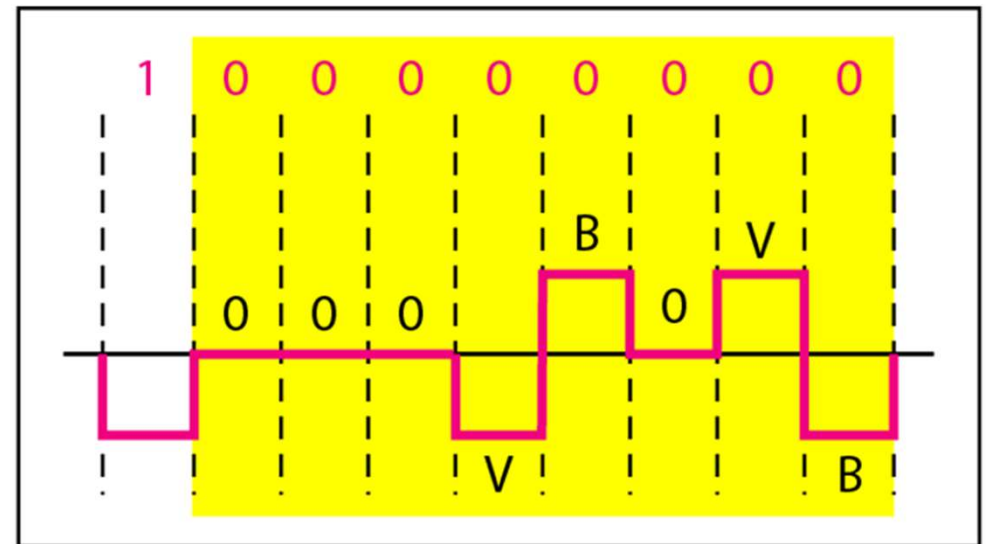
Scrambling

- B8ZS (Bipolar with 8 Zero Substitution)
 - Used mainly in North America
- 00000000 → 000VB0VB
- V (Violation): a non-zero level that violates AMI rule
- B (Bipolar): a non-zero level that obeys AMI rule

Scrambling



a. Previous level is positive.

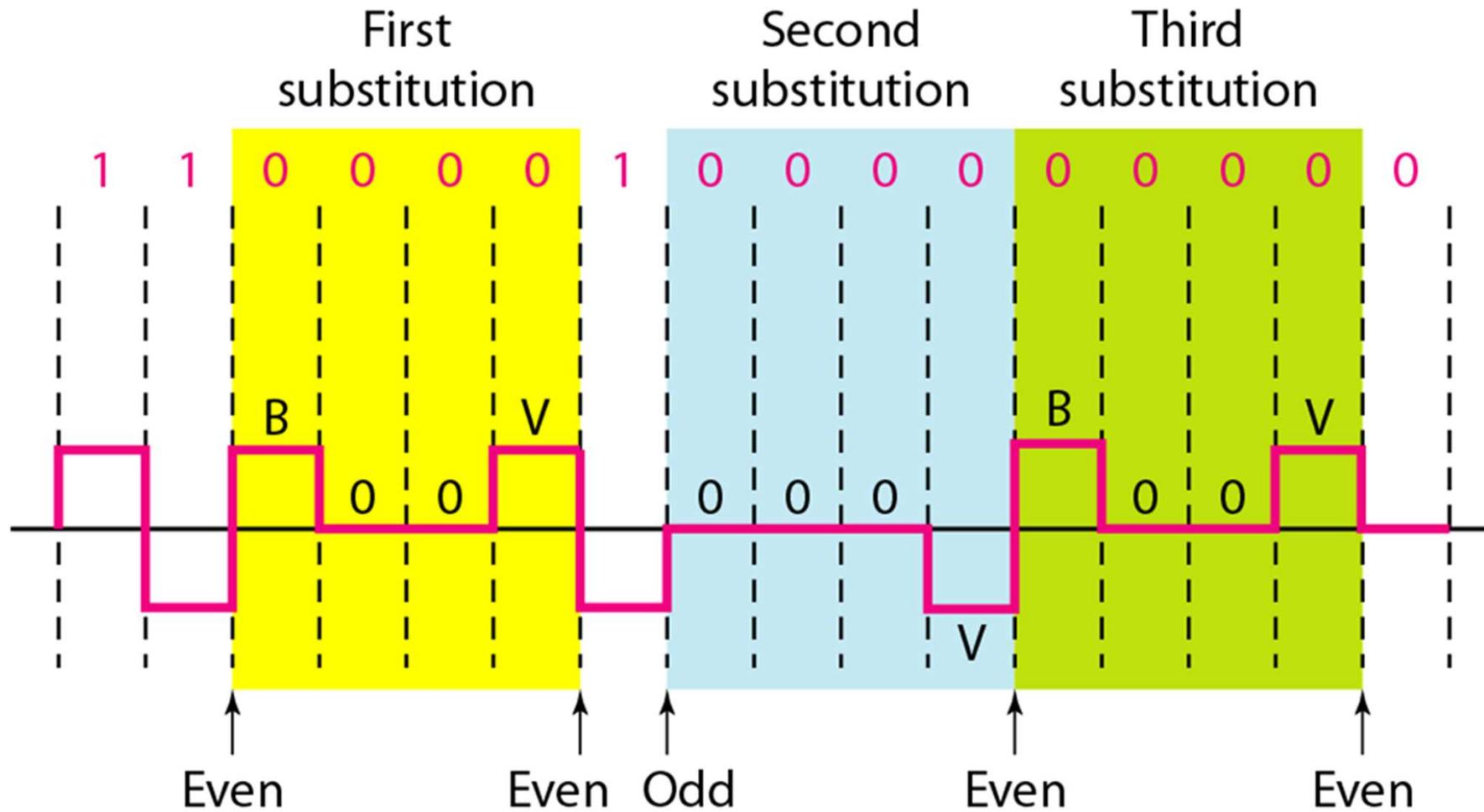


b. Previous level is negative.

Scrambling

- HDB3 (High-Density Bipolar 3-Zero)
 - Used in places other than North America
- 0000 $\rightarrow$ 000V, or 0000 $\rightarrow$ B00V
- After previous substitution, number of non-zero signal elements is odd $\rightarrow$ 000V
- After previous substitution, number of non-zero signal element is even $\rightarrow$ B00V

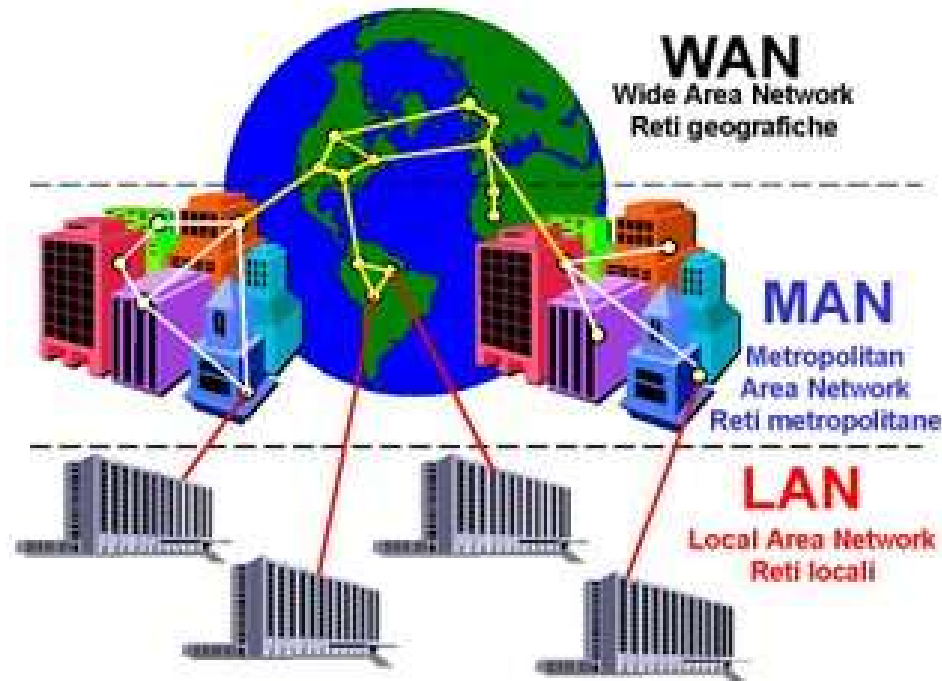
Scrambling



Case Study: Ethernet Physical Layer

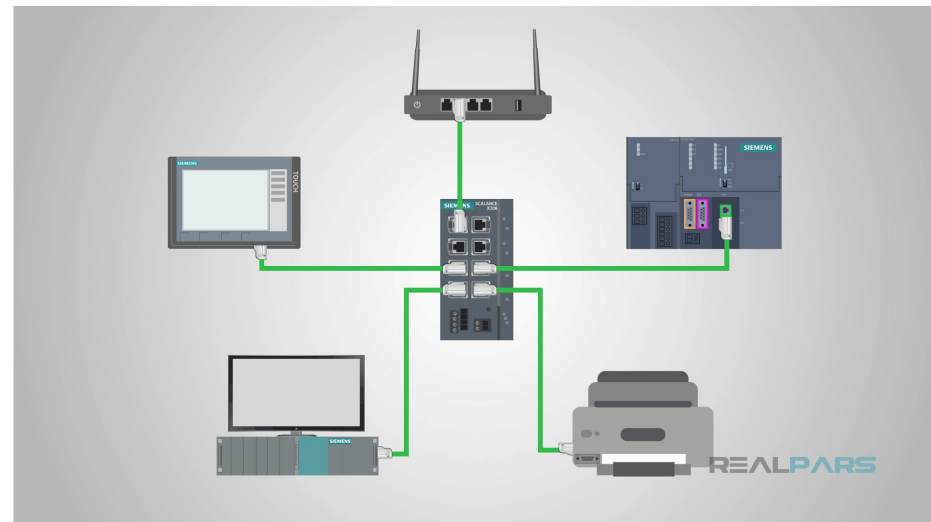
Ethernet

- A family of wired networking technologies commonly used in local area networks (LAN), metropolitan area networks (MAN), and wide area networks (WAN)

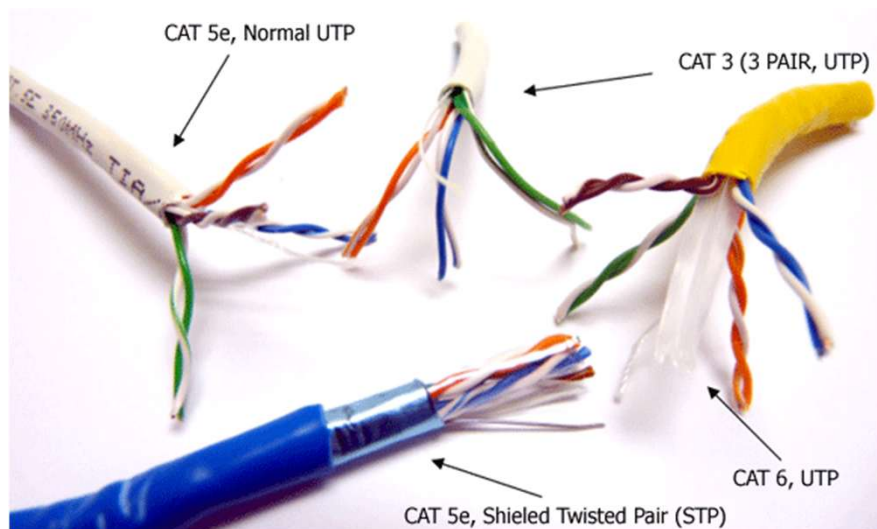


Ethernet

- Standardized by IEEE 802.3 division
 - Wi-Fi is IEEE 802.11
- Covers physical layer and data link layer



Ethernet Cable



칼론 CAT5 10M 랜선 랜케이블, 그레이, 1개



넥시 CAT.6 UTP 케이블 3m, 화이트, 1개

CATEGORY	SHIELDING	MAX TRANSMISSION SPEED (AT 100 METERS)	MAX BANDWIDTH
Cat 3	Unshielded	10 Mbps	16 MHz
Cat 5	Unshielded	10/100 Mbps	100 MHz
Cat 5e	Unshielded	1000 Mbps / 1 Gbps	100 MHz
Cat 6	Shielded or Unshielded	1000 Mbps / 1 Gbps	>250 MHz
Cat 6a	Shielded	10000 Mbps / 10 Gbps	500 MHz
Cat 7	Shielded	10000 Mbps / 10 Gbps	600 MHz
Cat 8	Shielded	25 Gbps or 40Gbps *	2000 MHz

Ethernet Standards

- Major standard variants
 - 10BASE5: 10Mbps, baseband transmission, max 500m
 - Uses RG-8 thick cable
 - 10BASE2: 10Mbps, baseband transmission, max 200m
 - Uses RG-8 thin cable
 - 10BASE-T: 10Mbps, baseband transmission, twisted pair
 - max segment length about 100m
 - 100BASE-TX: 100Mbps, full-duplex communication using two twisted pairs, called "Fast Ethernet"
 - 1000BASE-T: 1Gbps, uses 4 twisted pairs, 5-level coding
 - 10GBASE-T: 10Gbps

Ethernet Standards

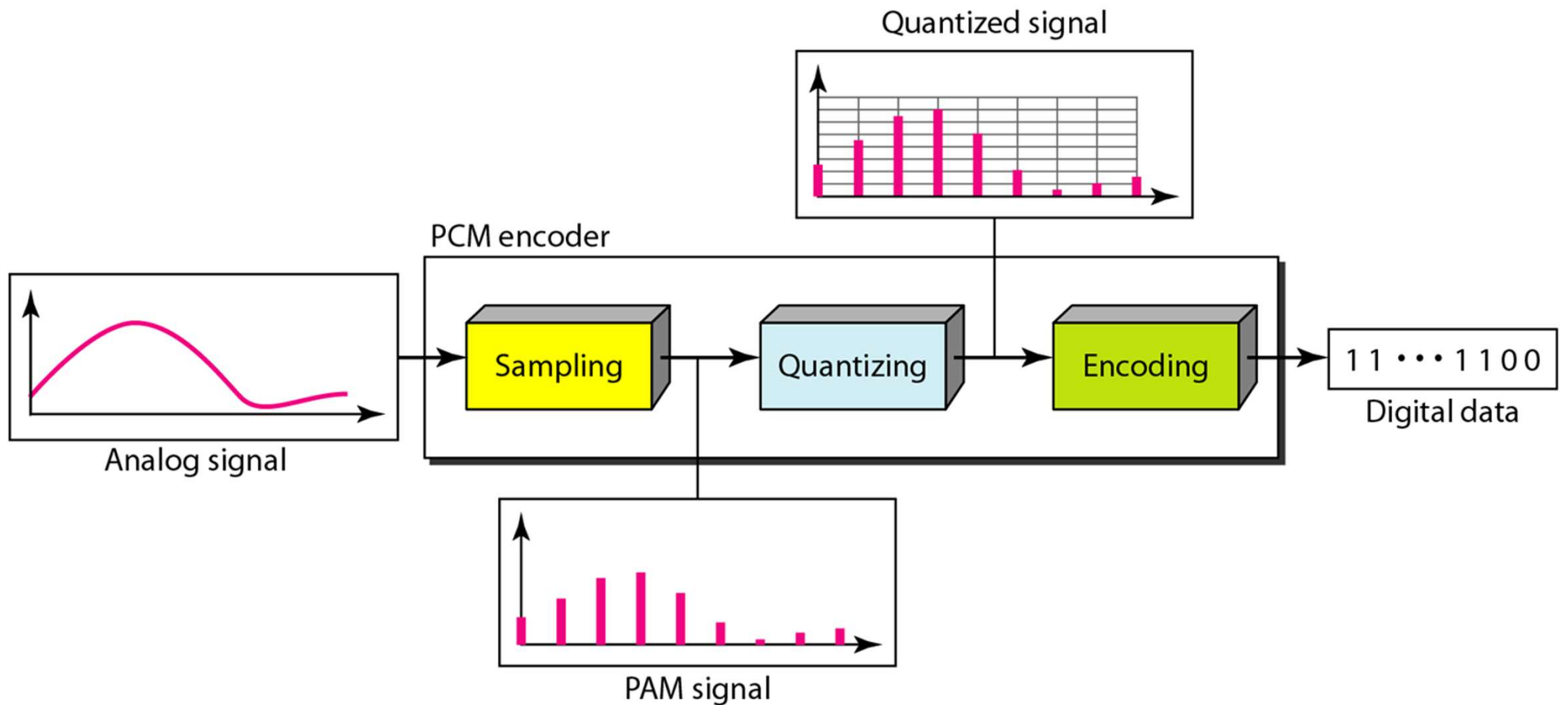
- Coding Schemes
 - 10BASE-5, 10BASE-2, and 10BASE-T use Manchester Encoding
 - 100BASE-TX uses MLT-3 encoding + 4B/5B
 - 1000BASE-T uses 8B1Q4 and 4D-PAM5
 - 10GBASE-T uses PAM-16 + DSQ128 + LDPC

Analog-to-Digital Conversion

Analog-to-Digital Conversion

- Multimedia data such as voice, music, movie, etc. are originally analog data
- In order to transmit and receive, the analog data should be converted to digital data

Pulse Code Modulation (PCM)

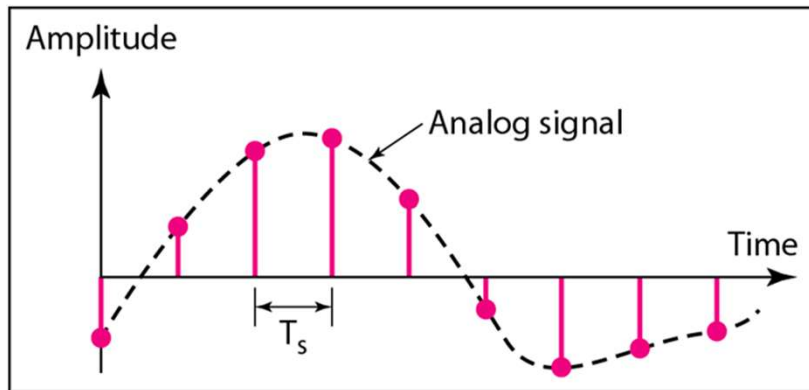


Pulse Code Modulation (PCM)

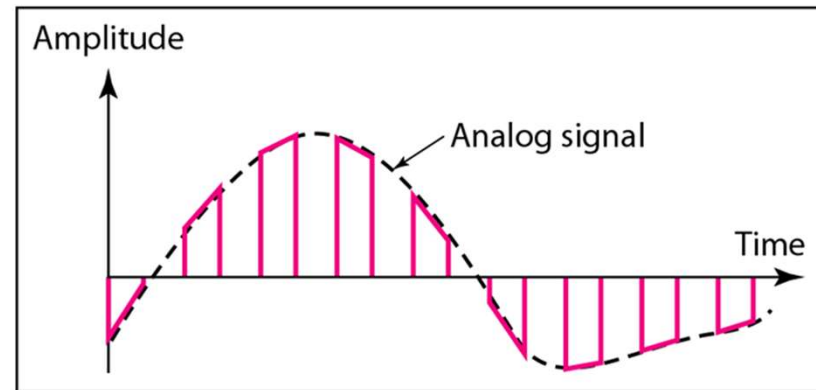
- The most basic analog-to-digital conversion scheme
 - Sample analog data
 - Quantize signals
 - Encode signals into digital bitstream

Sampling

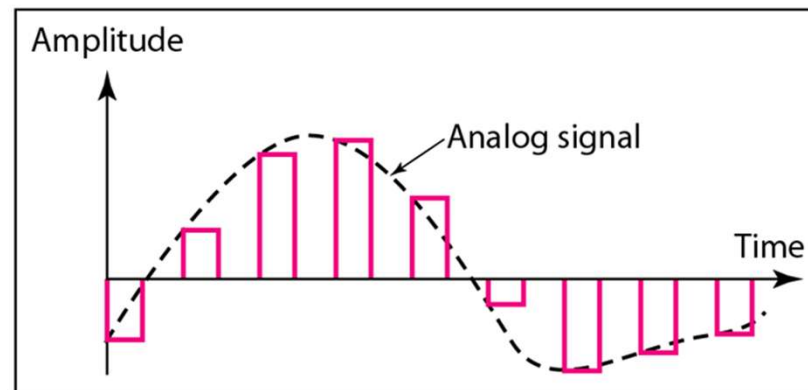
- Periodically read signal amplitude



a. Ideal sampling



b. Natural sampling



c. Flat-top sampling

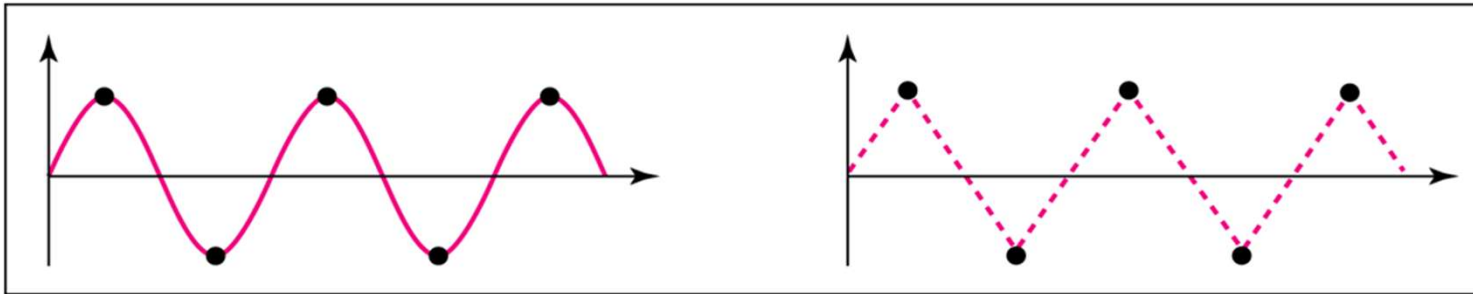
Sampling

- Sampling period
 - What is the period that the signal is sampled?
- Sampling rate
 - How many times the signal is sampled in a second
- Sampling period x sampling rate = 1

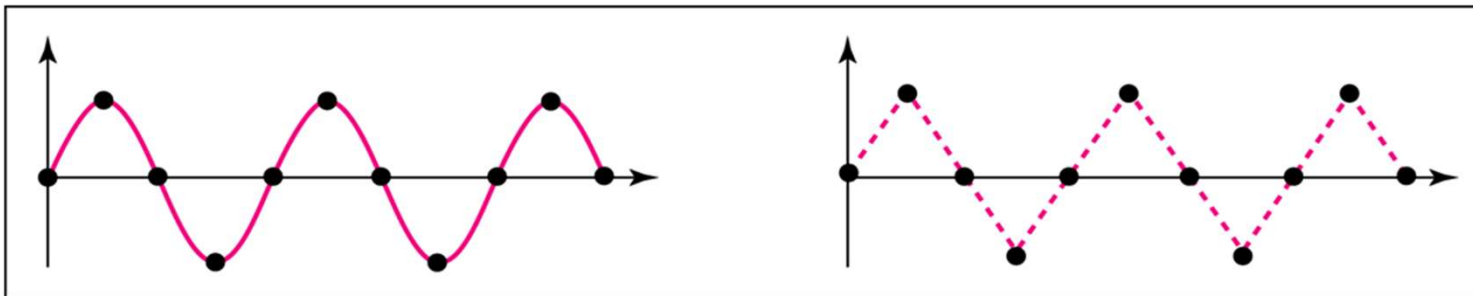
Sampling

- Nyquist theorem (Sampling theorem)
 - The sampling rate must be at least twice the maximum frequency of the signal
 - $f_s \geq 2 \times f_{max}$
 - If $f_s < 2 \times f_{max} \rightarrow$ aliasing (signal cannot be reconstructed)

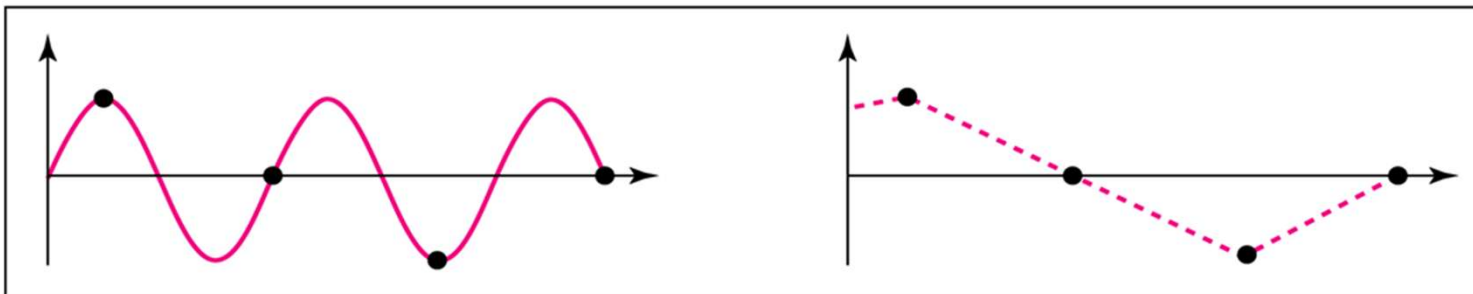
Sampling



a. Nyquist rate sampling: $f_s = 2 f$



b. Oversampling: $f_s = 4 f$



c. Undersampling: $f_s = f$

Example

- Telephone companies digitize voice by assuming a maximum frequency of 4000 Hz. What is the minimum sampling rate?

Example

- A complex low-pass signal has a bandwidth of 200kHz. What is the minimum sampling rate for this signal?

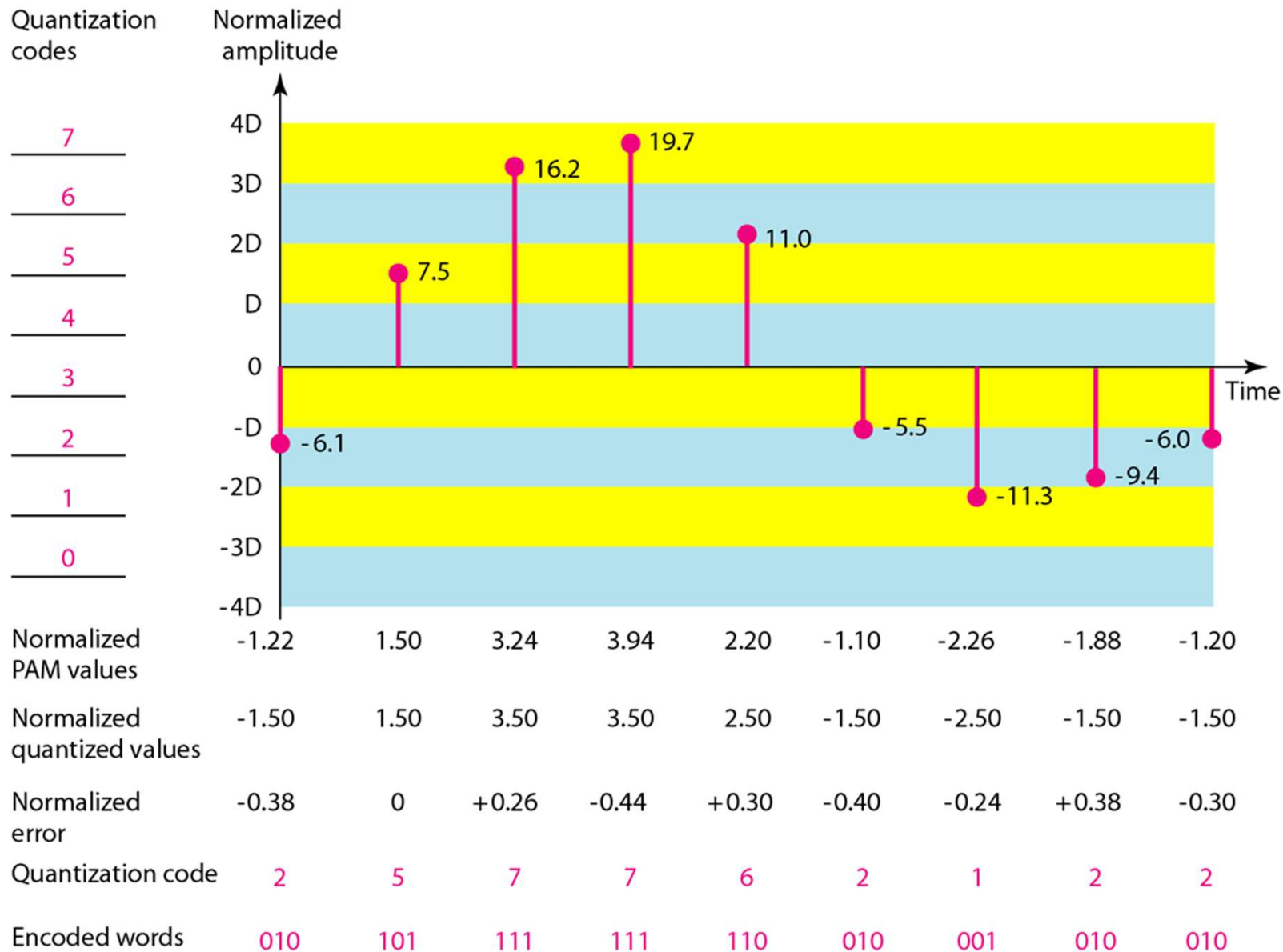
Quantizing

- Quantizing: map sampled amplitude values to a finite set of discrete levels
- The sampled signal level can be anywhere between minimum and maximum amplitude
- We need to quantize the signal into discrete signal levels in order to encode them into a digital bit stream

Quantizing

- Suppose the analog signal has amplitude between V_{min} and V_{max}
- Divide the range (V_{min}, V_{max}) into L sections of size D
- $D = (V_{max} - V_{min}) / L$
- Assign quantized code for each section

Quantizing



Quantizing

- Quantization Error
 - Difference between real value and quantized value
 - $-D/2 \leq \text{error} \leq D/2$
 - As number of quantization levels increases, quantization error decreases
- Quantization error affects SNR_{dB} as follows:
- $\text{SNR}_{\text{dB}} = 6.02n_b + 1.76\text{dB}$
 - n_b : bits per sample

Exercise

- What is the SNR_{dB} in the previous example?

Exercise

- A telephone subscriber line must have an SNR_{dB} above 40. What is the minimum number of bits per sample?

Encoding

- Convert quantized levels into bit patterns
- If there are L levels, $n_b = \log_2 L$

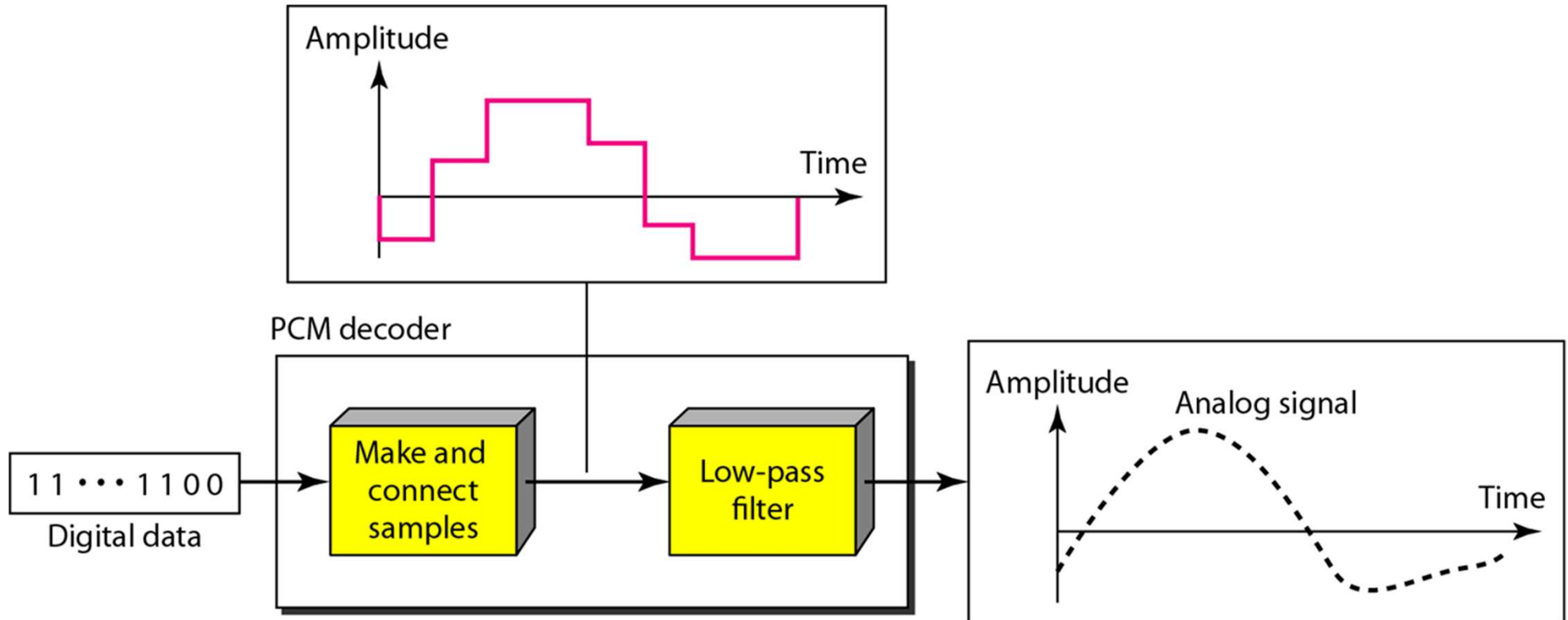
Encoding

- Unit: bit rate
- Bit rate = sampling rate $\times n_b$

Exercise

- We want to digitize the human voice. The frequency of human voice is between 0 and 4000Hz. What is the bit rate, assuming 8 bits per sample?

PCM Decoder



End of Chapter
